

LiveShowVR: A Tool to Navigate through Controlled Environments for Immersive Presentations

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Figure 1: Presenter's point of view. On the left side, the presenter is pointing at the sloth while activating the effects menu. On the right side, the presenter is activating the audience management and support menu.

ABSTRACT

Through the practical application of the subject “Exploring Costa Rica’s biodiversity through its national currency” and taking inspiration from special effects consoles found in theaters, the LiveShowVR tool allows the reimagination of traditional slides as immersive scenes. The presenter has subtly integrated controls at their disposal, with specific functionalities and previously set special effects for each scene. It ranges from basic needs such as note reviewing, microphone use, and audience management, to complete control over the visuo-haptic effects of their environment.

Index Terms: Virtual reality, user experience, virtual reality interfaces, presentation design, biodiversity.

1 INTRODUCTION

LiveShowVR tool transforms the conventional presentation model of 2D slides by utilizing the sensory capabilities of virtual reality to create immersive scenes with real special effects in 3D. The newly proposed interaction is directly inspired by live theater entertainment, which features real-time special effects, adaptable timing, and audience interaction. Based on these ideas, the designed user interface developed in Unity allows the presenter to become the showman of their own presentation, guiding the audience with dynamic effects and scene-specific interactions.

Given the speed and immediacy required in the task of creating a presentation, an efficient user interface was developed, which can be adapted to the story created by the presenter.

2 INTERFACE FEATURES

The prototype demonstrates the presenter’s interaction with the presentation using a fixed interface, below eye level (Fig. 2) and radial

menus accessed through the controllers (as shown on Fig. 1).

The interaction is guided through scenes (each one represented by an icon in the scene controller bar, as shown on Fig. 2). Each scene has different predefined effects and tools for controlling the experience. The interface shows the current and available scenes, audience questions, microphone status, and a timer (Fig. 2).

To provide an unobstructed view on each scene, the effects are activated via controller menus. Also, features related to audience interaction, including the ability to receive questions, object interaction, and environment exploring are also included here.

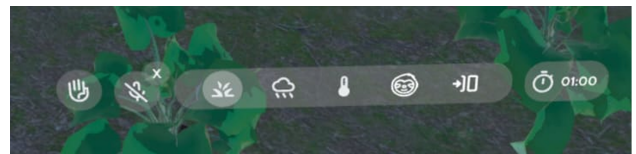


Figure 2: Fixed interface. From left to right: question requests display, on/off microphone status, scene controller bar, timer display.



Figure 3: On the left, Costa Rican bills. On the right, inspired environments to showcase the presentation theme.

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3 ENVIRONMENT

The experience was developed in Unity, using three environments created in Blender to demonstrate the flexibility of the user interface. These were designed to follow the chosen theme: “Exploring Costa Rica’s biodiversity through its national currency.” Each environment consists of themed rooms intended to showcase the capabilities of the tool in terms of sensory effects. It includes a museum room (the lobby), a rainforest room and a coral reef room, these last two inspired by ₡10 000 and ₡2000 Costa Rican bills. (As seen on Fig. 3).

4 CONTROLLER-ASSIGNED ACTIONS

Both left and right controllers are used to interact with the interface and environment. Fig. 4 shows the assigned action to each button for the left and right controller, which are visible for users while using the application. The original letter button icons (A, B, X, Y) were modified to display an icon for their respective functionality.

An effort was made to mirror the interactions across both controllers so that the main Grab and Trigger buttons would have consistent functions (option menus and selection), allowing intuitive use of both controllers with distinct, yet complementary roles for each hand (as recommended by [1]). The left controller is focused on audience control, while the right controller is focused on scene changes and effects. (Fig. 5).

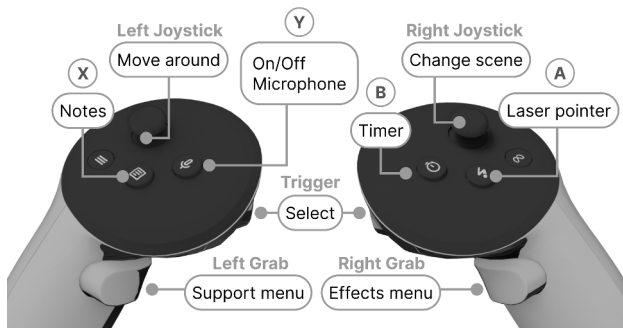


Figure 4: On-screen controllers display and user interface actions assigned to each button.



Figure 5: Controller menus displayed while pressing the left or right Grab button. On the left: support menu. On the right: effects menu.

5 MENU DESIGN

The interface design was iteratively refined through multiple prototype versions. Early designs explored more persistent on-screen elements, which were later simplified based on usability testing, resulting in the controller-based interaction model.

The options menu is designed as an up-to-eight-options radial menu (Fig. 5) that is displayed only while the Grab button is being held. Prior work has shown that, compared to linear menus, this

approach reduces target seek time and selection errors when the number of items is limited, as seen on [2].

The menu appears around the controller when the Grab button is pressed and remains fixed in that position as long as the button keeps being held. The user then moves the controller towards the desired option and releases the button to select it (Fig. 6).



Figure 6: Activating the text effect. On the left side the presenter is choosing the text option on the effects menu and, on the right, the predetermined text is displayed.

6 AUDIENCE MANAGEMENT AND SUPPORT MENU

The presenter can access the support menu by using the Grab button on the left controller. These options are permanent and consist of tools that are accessible to the presenter throughout the entire presentation, with the purpose of providing flexible experience for the audience. The current tools are: Home (allows the presenter to go back to the lobby), Receive/Stop Questions (enable/disable questions from the audience), and Wander/Gather (enables the audience to move freely around the scene or sets the position of the audience back towards the presenter).

7 SCENE-PREDETERMINED EFFECTS

The presenter can access the effects menu by using the Grab button on the right controller. The preset options include effects that can be applied to the environment depending on each scene. It is important to note that the presenter has full control over the timing and duration of each effect, allowing the presentation to be managed as desired. The effects presented in this demo include: Sound effects, haptic controller effects, ambience control (weather), insert non-interactive animations, insert and stop interactive/audience-shareable objects, and environment switching.

8 RESULTS AND FUTURE WORK

The proposed system employs a user interface that enables presenters to deliver live, effect-rich stories within a virtual environment. Current implementation supports only the presenter’s point of view during a preconfigured presentation. Future work will focus on the development of a spectator interface, incorporating interaction mechanisms between the audience, the presenter, and the virtual environment.

As a potential long-term direction, the development of an authoring tool for scene creation and preset customization could be explored, enabling adaptability to different presentation contexts, themes, and user requirements.

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